

Purpose

Hypothesis testing evaluates whether sample evidence is strong enough to challenge a claim about a population parameter. Every test follows the same high-level structure: state hypotheses, choose a significance level, compute a test statistic, and compare it to a rejection threshold (critical region or p-value).

Hypotheses

- **Null (H_0):** Represents "no effect" or the status quo; always includes equality ($=, \leq, \geq$).
- **Alternative (H_1):** Represents the research claim; can be two-tailed ($\neq$), left-tailed ($<$), or right-tailed ($>$).

Critical Regions & Tail Types

- **Two-tailed:** Shaded regions on both ends of the curve ($\pm$ critical); reject H_0 when statistic $\leq -t^*$ or $\geq +t^*$.
- **Left-tailed:** Shaded region on the left; reject H_0 when statistic $\leq$ critical value.
- **Right-tailed:** Shaded region on the right; reject H_0 when statistic $\geq$ critical value.

Alt-text: Each diagram shows a bell curve with shaded rejection areas corresponding to the row description.

Significance Level (α)

α is the probability of committing a Type I error. Common choices: 0.10, 0.05, 0.01. Split α across tails for two-tailed tests.

Test Statistic

General form: $(\text{estimate} - \text{hypothesized value}) / \text{standard error}$. The distribution (z, t, χ^2, F) depends on the test, sample size, and assumptions.

Decision Approaches

1. **Critical Value Approach:** Compute the critical value(s) using α and df; reject if statistic falls in rejection region.
2. **p - Value Approach:** Compute p - value; reject if p - value $< \alpha$.

Workflow

1. Translate the claim into H_0/H_1 .
2. Set α and determine tail direction.
3. Check assumptions (randomness, independence, distribution requirements).
4. Gather data and compute descriptive stats.
5. Calculate the test statistic.
6. Find critical value(s) or p - value.

7. State decision (reject / fail to reject) and interpret in context.

8. Pair with a confidence interval when possible for estimation insight.

Related Resources

- See `type1\_type2\_errors.pdf` for error definitions.
- See `test\_power\_and\_examples.pdf` for power analysis.
- Consult per-topic how-to PDFs for assumptions and formulas.